

SHORT WAVE

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DEFINITION OF SHORT WAVE

- Shortwave means the application of high frequency electrical energy to the body tissues in order to bring physiological and therapeutic effects, these effects can be achieved by thermal or non thermal effect.



- **There Are Two Circuits Here:**

1. Machine circuit : composed of a high frequency generator, amplifier, oscillator coil.
2. Patient circuit: composed of a resonator coil, variable capacitor, electrodes and the treated tissue.





PRZYKŁADY ZASTOSOWAŃ













SHORTWAVE PRODUCES BOTH ELECTRIC AND MAGNETIC FIELDS.

THE AMOUNT OF ELECTRIC FIELD TO MAGNETIC FIELD IS DEPENDENT ON:



1. The characteristic of the machine itself (dependent on the manufacturer).
2. The type of electrodes that we use.
3. Shortwave Can Be Used in Two Modes:

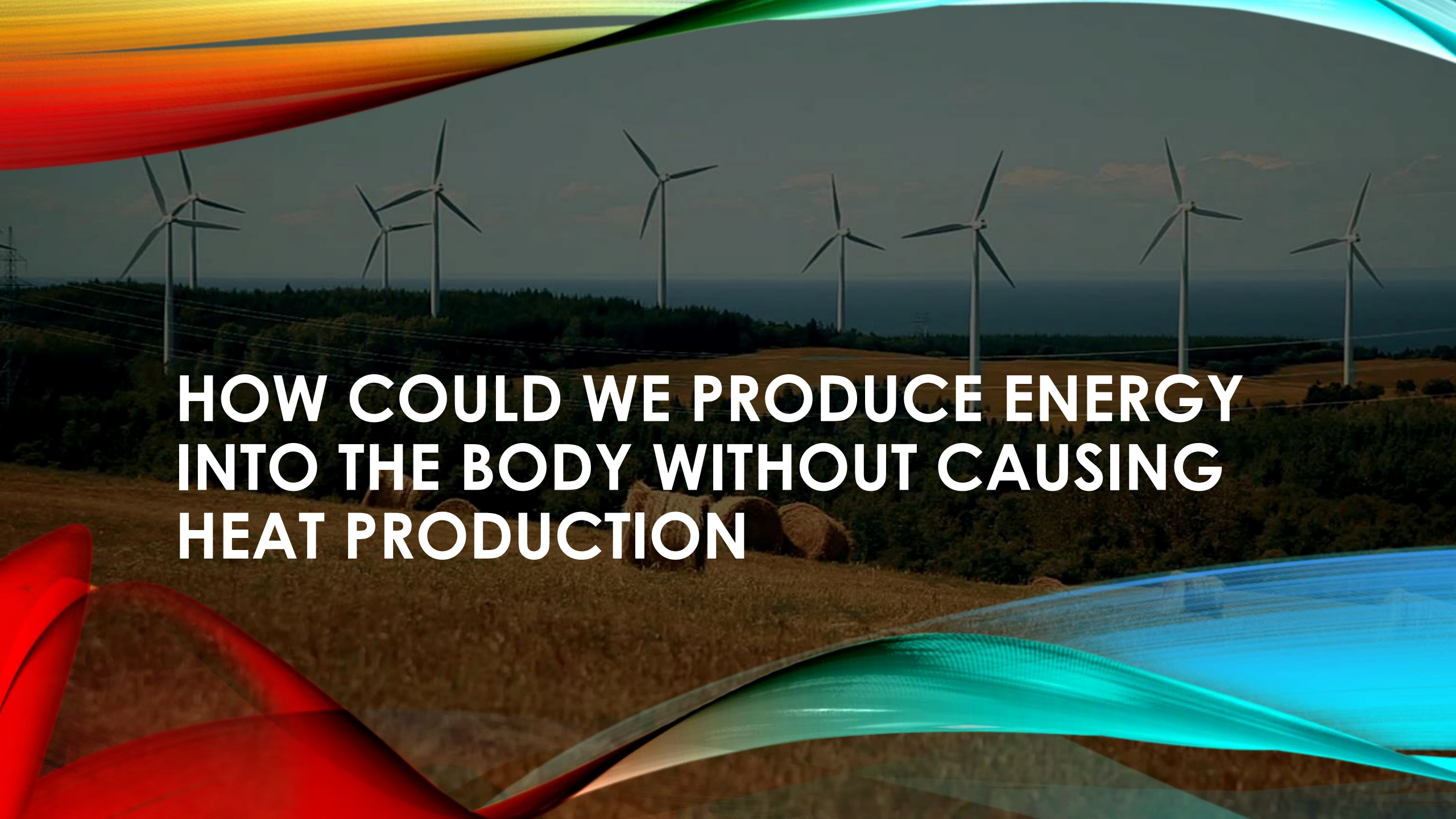
- Continuous (energy is emitted all the time).
- Pulsed (energy is emitted part of the time).



When using the pulsed shortwave, there are certain parameters that we need to know:

Pulse width: measured by micro seconds.

Pulse frequency: how many pulses we have per second measured by Hz or pulses per second (PPS)

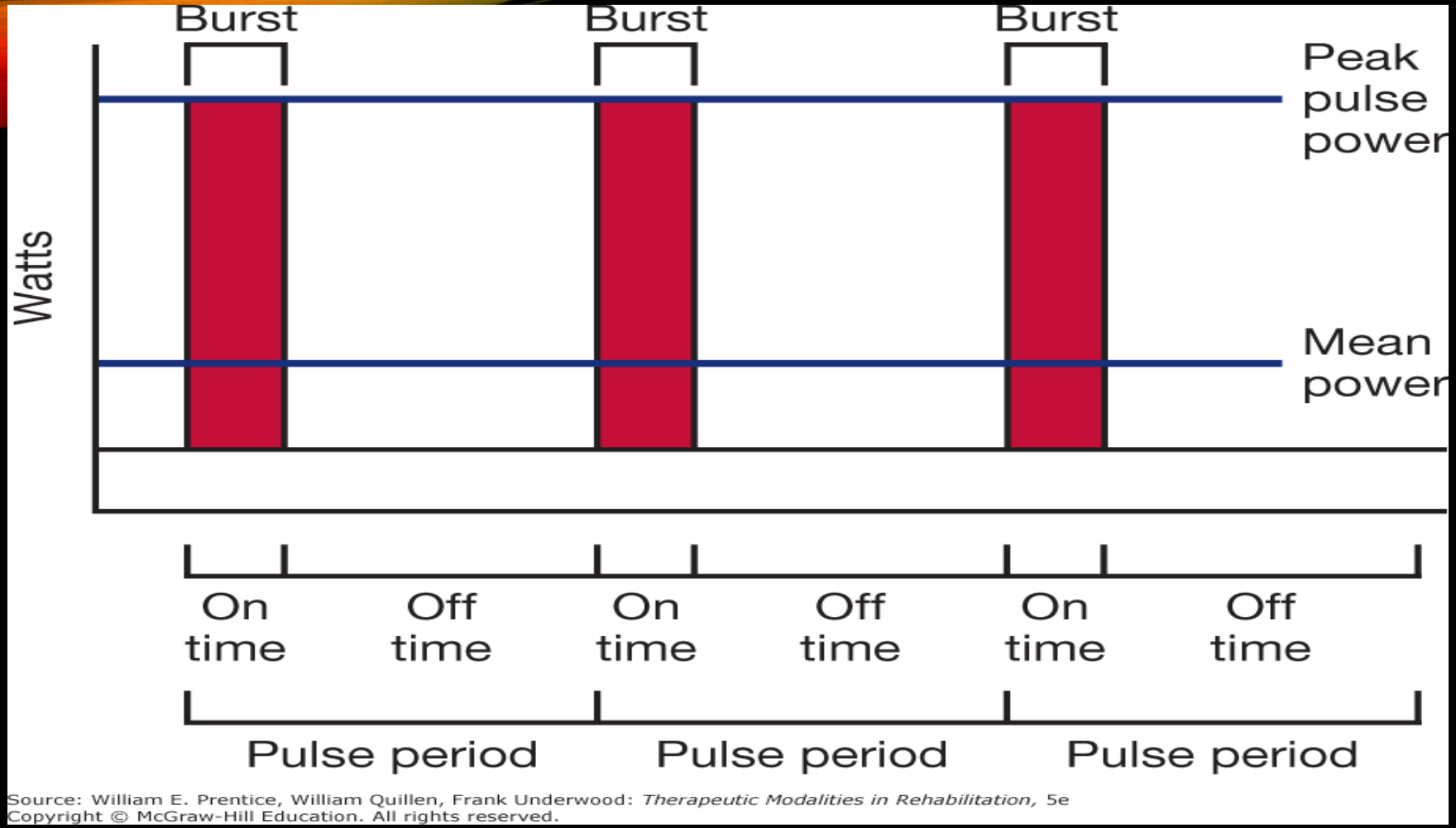


HOW COULD WE PRODUCE ENERGY INTO THE BODY WITHOUT CAUSING HEAT PRODUCTION

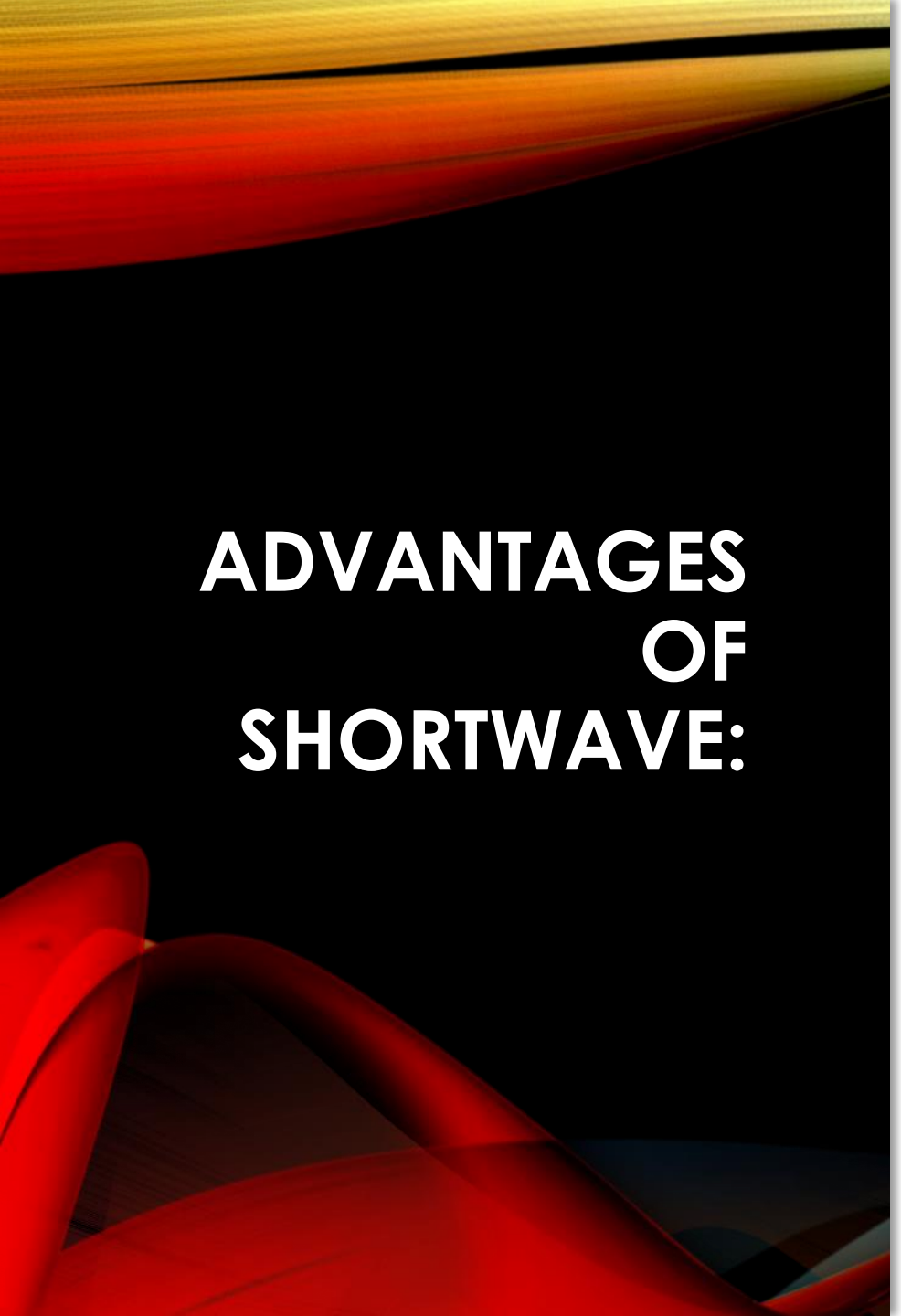
PULSED SHORT WAVE

- The off time with pulsed shortwave is longer than the on time which allows the dissipation of heat.
- The interpulse period gives time for the heat to dissipate.
- If the interpulse is short, heat will develop, because there is no time for the effect to dissipate within the body.
- Pulsed shortwave can only be non thermal if short pulses, low pulse repetition rate and short treatment duration is being used.





Source: William E. Prentice, William Quillen, Frank Underwood: *Therapeutic Modalities in Rehabilitation,* 5e
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ADVANTAGES OF SHORTWAVE:

- The deep penetration, it goes deep to 5cm.
- You can use it while attending to another patient, which allows you to manage your time in the clinic.
- It covers large areas.
- It can heat the deeper tissue without heating the superficial tissue
- It can be used with sensitive tissues because it doesn’t need to be in contact with the treated tissue.

Primary Effects of PSWD

1. Increased number of white cells, fibroblasts etc in a wound.
2. Improved rate of edema dispersion.
3. Encourages absorption of heamatoma.
4. Reduction *(resolution)* of the inflammatory process.
5. Prompts a more rapid rate of orientation & deposition of collagen.
6. Encourages collagen layering at an early stage.
7. Stimulation of osteogenesis.
8. Improved healing of the peripheral & central nervous systems

Sreeraj S R

Effects on Tissue

- **Skin & Subcutaneous Tissue:** Mild warming; increased capillary permeability
- **Muscle:** Relaxation of spasm; ↑ extensibility; reduced muscle viscosity
- **Connective Tissue:** Elastic collagen fibers become more pliable; improved range of motion
- **Bone & Cartilage:** Limited direct heating; however, surrounding tissue warming can aid joint mobility
- **Cellular Level:** PSWD promotes ATP synthesis, protein synthesis, and mitotic activity—key for tissue repair

Why Use PSWD?

Superficial Heat Pack (Trowbridge et al 2004)	Ultrasound (Garrett et al 2000)	PSWD (Garrett et al 2000)
Increased intramuscular temperature 1.0°C at 2 cm deep	Increased intramuscular temperature 0.17°C at 3 cm deep	Increased intramuscular temperature 4.58°C at 3 cm deep
	Treatment size was the same as the PSWD treatment size	Treatment size was the size of the diathermy drum
	Returned to baseline temperature at 14.88 min	Returned to baseline temperature at 38.5 min

- Studies have shown PSWD increases intramuscular temperature about 4.0°C during treatment and decays about 1.8°C 10min post-treatment (Draper et al 1999; Draper et al 1997; Castel et al 1997)

Mechanism of Effect

ULTRASOUND

Dense collagen based tissues

- Ligament
- Tendon
- Fascia
- Joint capsule
- Scar tissue

PULSED SHORTWAVE

Wet, ionic, low impedance tissues

- Muscle
- Nerve
- Areas of oedema, haematomas and effusion

LASER

Superficial Vascular Tissues

- Open wounds
- Muscle
- Nerve
- Tendon sheath . .

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TYPES OF ELECTRODES:

1) Flexible plates
2) Space electrode
3) Monode
4) Minod
5) Diplode
6) Coil

Flexible pad	Space plate	Minod -monode	diplode	coil
				



METHODS OF SHORTWAVE APPLICATION

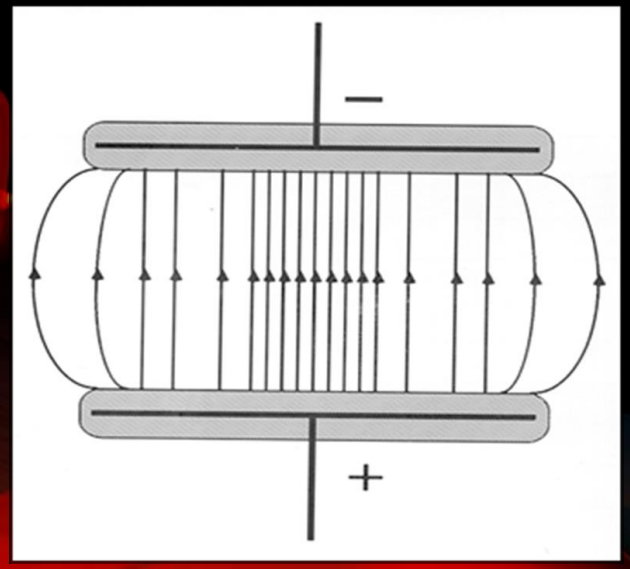
*Capacitive method.
* Inductive method

*CAPACITIVE METHOD.

Air space plates

Pad electrodes.

CAPACITANCE SWD



Places the patient within the actual circuit of the machine's energy.

Creates a stronger electrical field.

Produces more heating in the superficial tissues (adipose tissue) and bone/muscle interface.

As the current flows through the tissue, it causes oscillation of charged particle and thus increases tissue temperature.

CAPACITANCE SWD



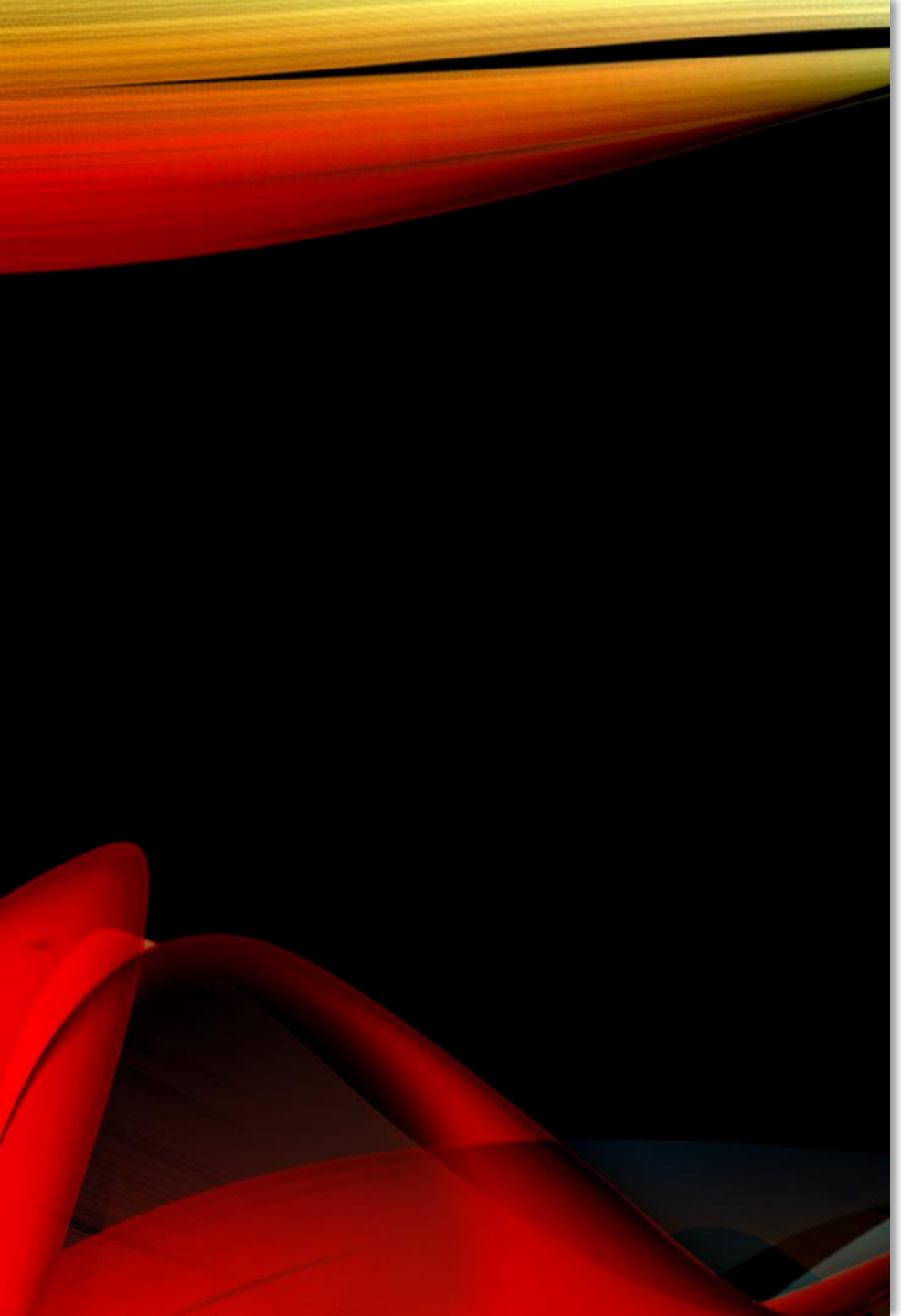
AIR SPACE PLATES:

They come in different sizes.

Can be used in almost every part of the body, but because they are rigid they will not fit every contour.

It can be put on either side of the limb, or can be used in the same side.

If the electrodes are placed on the same side there should be a distance between the electrodes, because if there is no distance there would be a shock, so we need to put a distance equal the diameter of the electrode.



Skin-electrode distance: is the distance between the electrode and the skin.

If we place to electrodes at the same distance there will be even heating.

If we placed one electrode closer to the skin than the other, the heating won't be even, it will be more near the closer electrode (superficial heat).

If we used a large electrode and a small one the heating will be more near the small (deep heat).



PAD ELECTRODES:

They are malleable metal covered with a thin layer of rubber

They must have uniform contact pressure on the body

Towels are used to prevent burns

When using them you should insure it is in full contact.

We have to rap it in towel that is 2 cm thick, less than that could cause a burn, more than that will affect the amount of heating created in the body.

Distance between the electrodes should be equal the diameter of the electrode, if it is less than that there will be a shock.

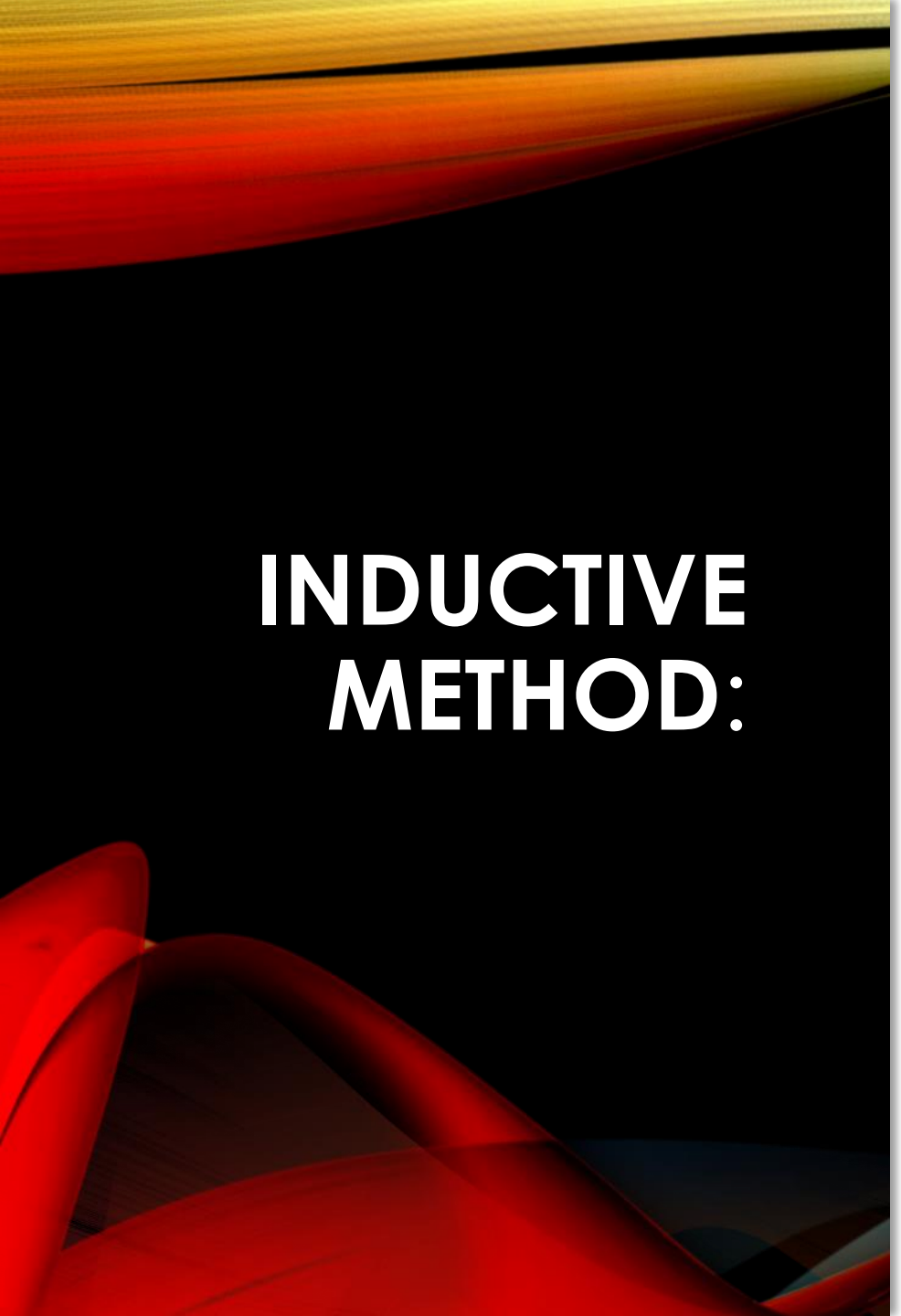
PAD ELECTRODES



Source: Prentice WE, Quillen WS, Underwood F: Therapeutic Modalities in Rehabilitation, 4thEdition: www.accessphysiotherapy.com

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INDUCTIVE METHOD:

A coil is housed within a drum

It is known to heat the deeper tissue without heating the superficial tissue.

Penetration to 3 - 5cm.

Make sure that the fat layer of the patient is not more than 2cm, because the penetration will be limited and the absorption of heat will accumulate in the fat layer which might cause a burn.



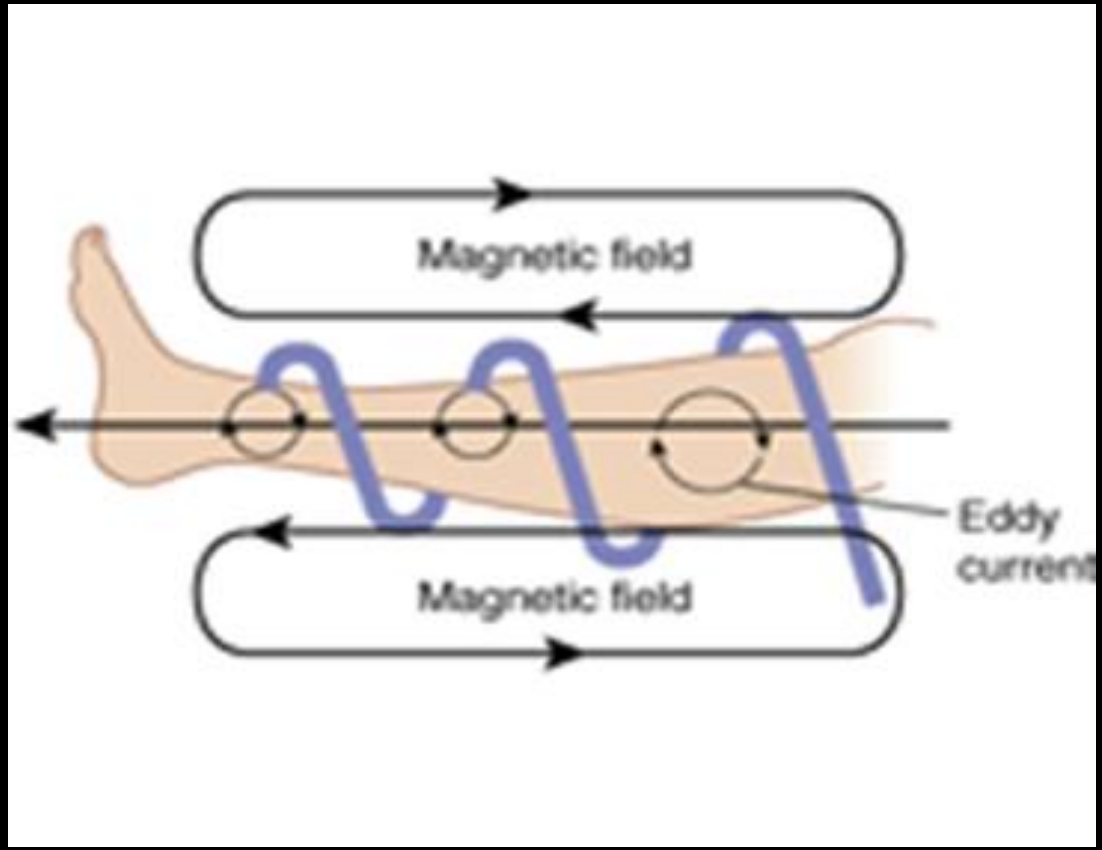
The tissues that are heated are blood vessels, muscles and sweat.

Current flowing within the coil produces a rotating magnetic field

Magnetic field produces eddy currents in the tissues

Eddy currents cause friction that produce heat

- Usually used for limbs
- Used in two ways:
  - wrapped around the part being treated.
  - Pancake (usually used on the back).

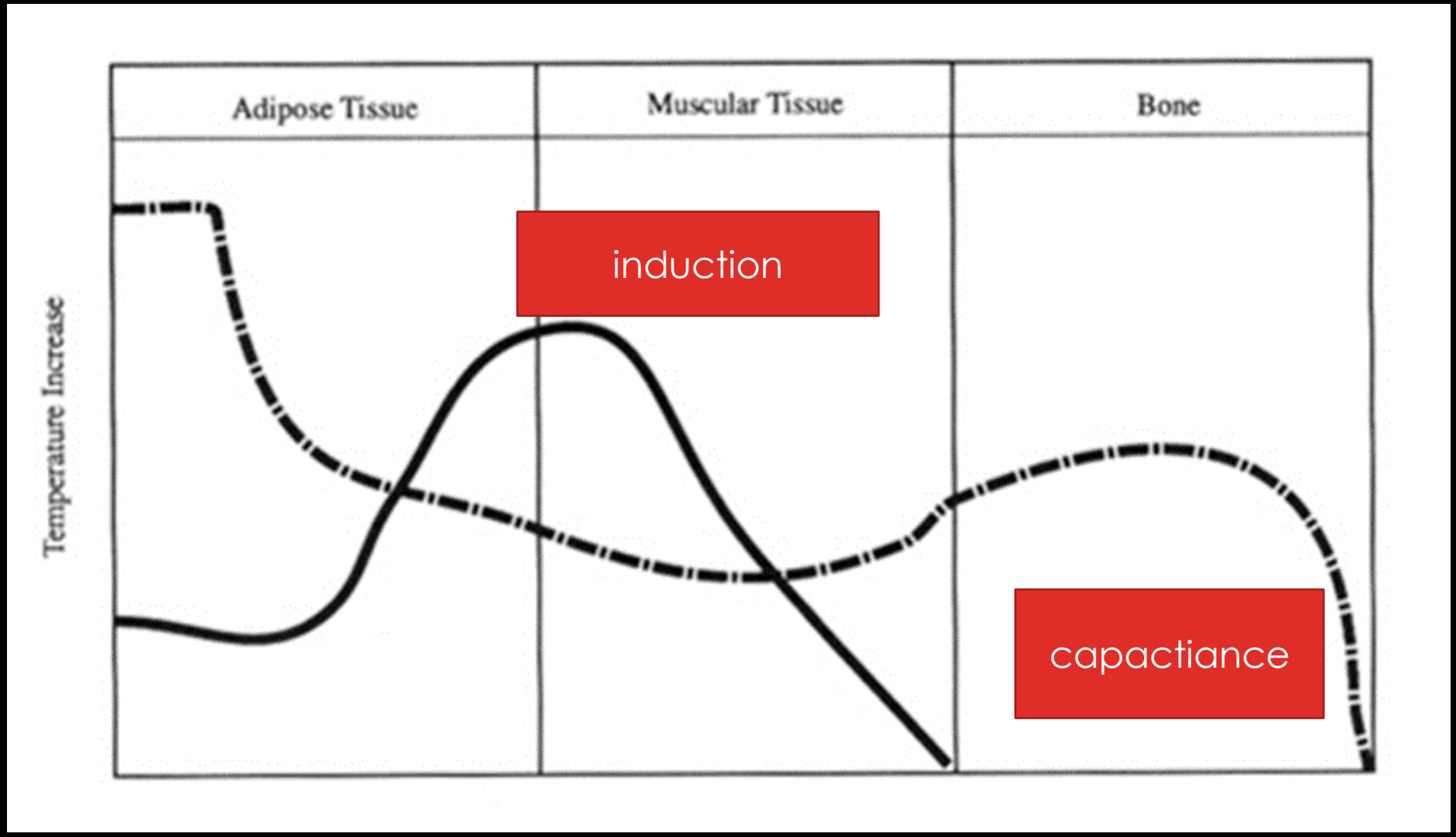


A PANCAKE APPLICATION OF INDUCTIVE SW



- When wrapping, the space between one turn and the other should be 5 to 10cm and it is insure by felt spacers (comes from the manufacturer).
- It is known to heat the deeper tissue without heating the superficial tissue.
- The tissues are placed in the electromagnetic field by the use of an insulated cable or coil electrode which is available in two forms Cable Electrodes and Drum Electrodes.

CAPACITANCE VS. INDUCTION



*Capacitive application produces more heating in the superficial tissues (adipose tissue) and bone/muscle interface while Induction application produces more heating in muscular tissue. **Why?**

- **SWD applicators differ in how they deliver energy—and thus where that energy concentrates—based on tissue conductivity (σ) and impedance (Z).**

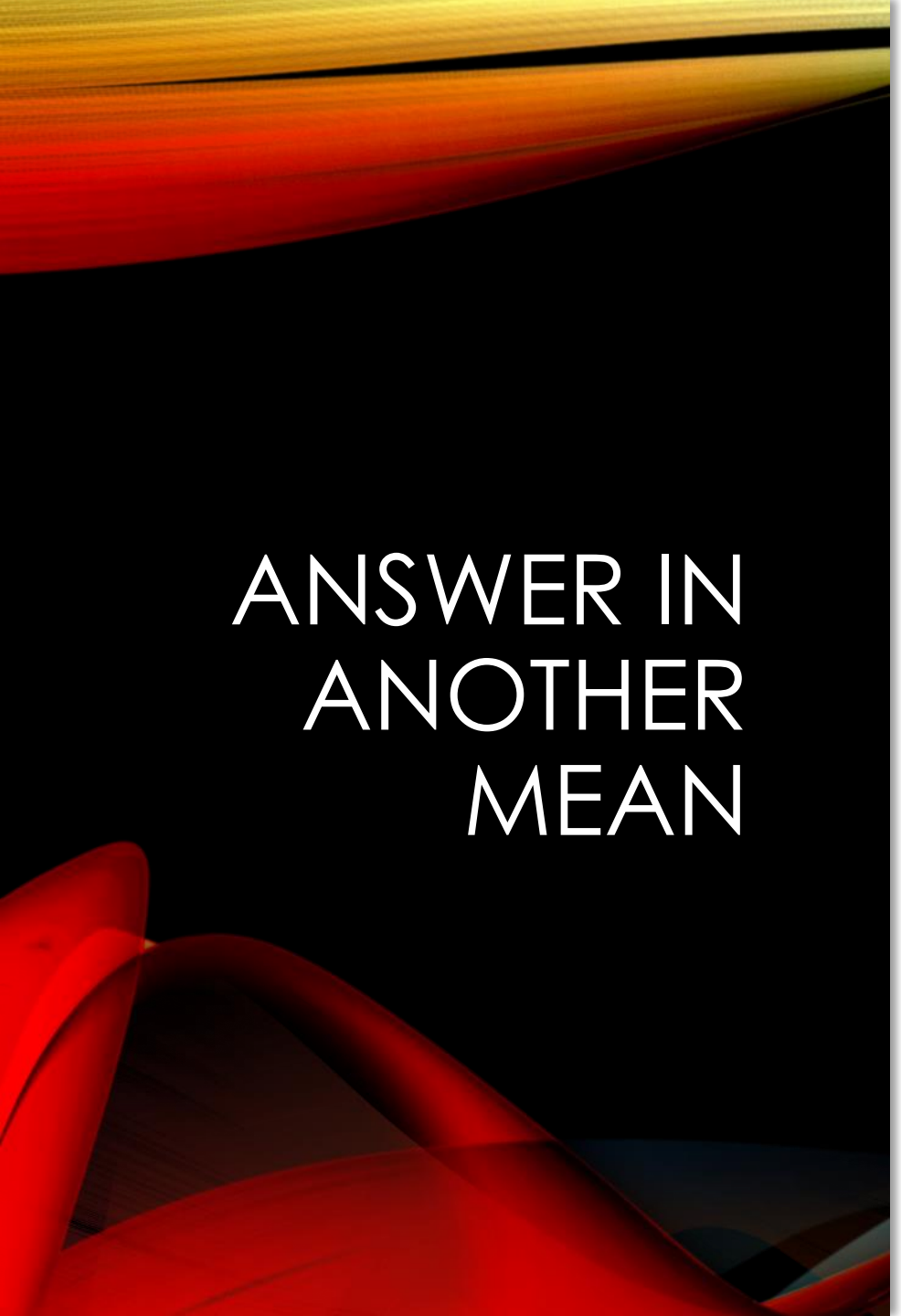
1. Capacitive (Plate-to-Plate) Application

1. *Mechanism:* Two opposing metal plates create an oscillating **electric field**. Current flows from one plate, through the patient's tissues, to the other plate.
2. *Heat Generation:*
   - $P = I^2 \cdot R$ (Joule's law). Low-conductivity tissues (i.e., **adipose**, high R) force higher voltage drops → greater local heating.
   - At interfaces (e.g., **bone/muscle**), abrupt changes in σ cause concentration of the electric field → "hot spots."
3. *Depth Profile:*
   - Most energy is absorbed in the first few centimeters, especially in fat layers and at tissue boundaries.

- Induction (Drum or Coil) Application
  - *Mechanism:* A coil produces a time-varying **magnetic field** that penetrates tissues unimpeded. This induces circular eddy currents in conductive tissues.
  - *Heat Generation:*
    • Eddy currents preferentially flow where σ is highest—namely **muscle** (high water/ion content).
    • Lower resistance means more current but less voltage drop; because of the large current loops, total heating remains greatest in muscle.
  - *Depth Profile:*
    • Magnetic fields are not shunted by superficial fat; they penetrate more uniformly, focusing heating in deeper, well-perfused musculature.

TAKE A MESSAGE

- Choose capacitive SWD for superficial heating (e.g., adipose-rich regions, bone/muscle interfaces) and inductive SWD when you want to target deep, high-conductivity muscle tissue.



ANSWER IN ANOTHER MEAN

The strength of magnetic field is affected by the distance of tissue from the applicator according to inverse square law.

The strength of the induced eddy currents is determined by:

- Strength of the magnetic field in the area

- Electrical conductivity of the tissue in the area.

- **The electrical conductivity of tissue depends primarily on**
- 1- Tissue type. 2- Frequency of the signal being applied.
- Metals and tissues with a high water and electrolyte content, such as **muscle or synovial fluid**, have high electrical conductivity, whereas tissues with low water content, such as **fat, bone, and collagen**, have low electrical conductivity.
- Thus, inductive coils can heat both deep and superficial tissues, but they produce the most heat in tissues with the highest electrical conductivity.

- the dielectric characteristics of tissues can also cause different temperature variations in them, since they offer variable resistance to the current passage. Thus, tissues like the adipose offer a high resistance to the current passage, and present a superior heating when compared to the muscular tissue, which due to its greater amount of electrolytes, presents a lower resistance to the radiofrequency currents.

METHODS OF APPLYING THE ELECTRODES:

Contraplanar electrodes are placed on opposite sides of the part, to treat deeply placed structures.

Coplanar: electrodes are placed on the same side of the part to treat more superficial structures.

Cross-fire: half the treatment is given with the electrodes in a diagonal way, the other half of the treatment is put on the opposite way. This method is usually used for sinuses and body cavities such as the uterus.

Longitudinal: one flexible electrode is used with a space electrode.



Photo (1) contraplaner technique



Photo (2) longitudinal technique





Photo (3) contraplanner technique

Co planer technique

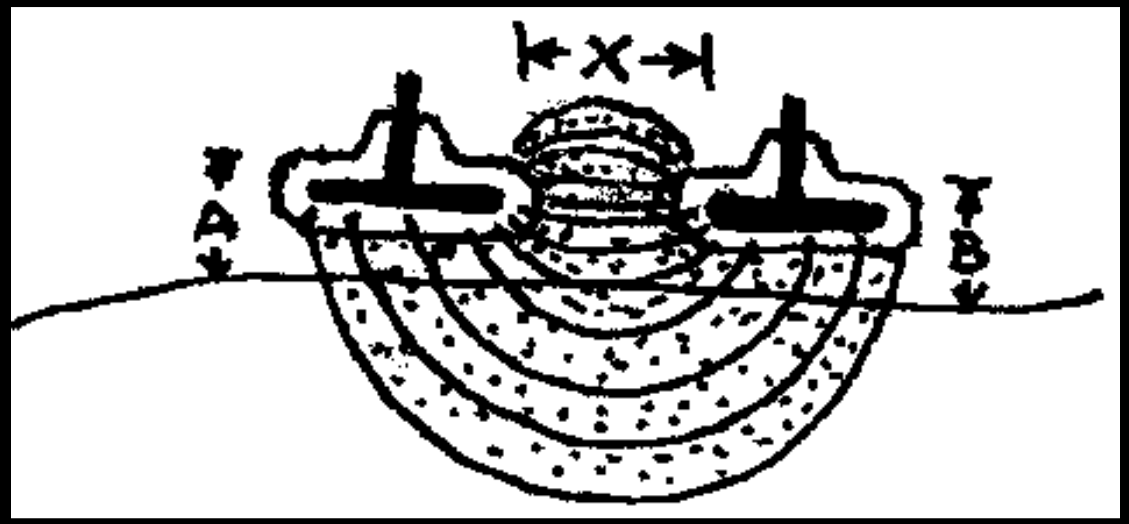


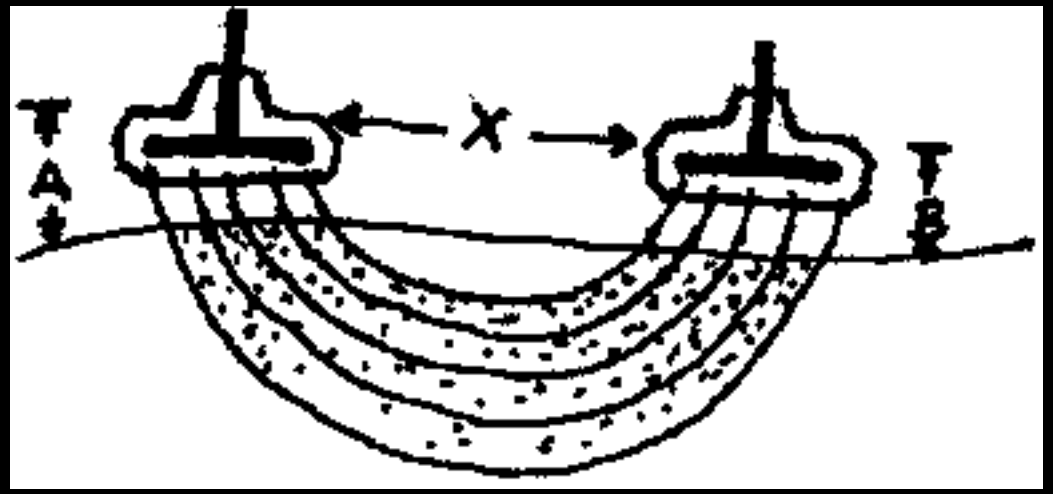
FACTORS INFLUENCING THE FIELD:

- 1-spacing of electrode:
- -Normal spacing of electrodes → (2-4cm) uniform field.
- -Closely spaced electrodes → superficial heating
- -Far spaced electrode → deep heating
- -Unequal spacing of electrode → more heat concentrated under closer electrode.

Short distance superficial heat will be.

Long distance deep heat will be.





FACTORS INFLUENCING THE FIELD:

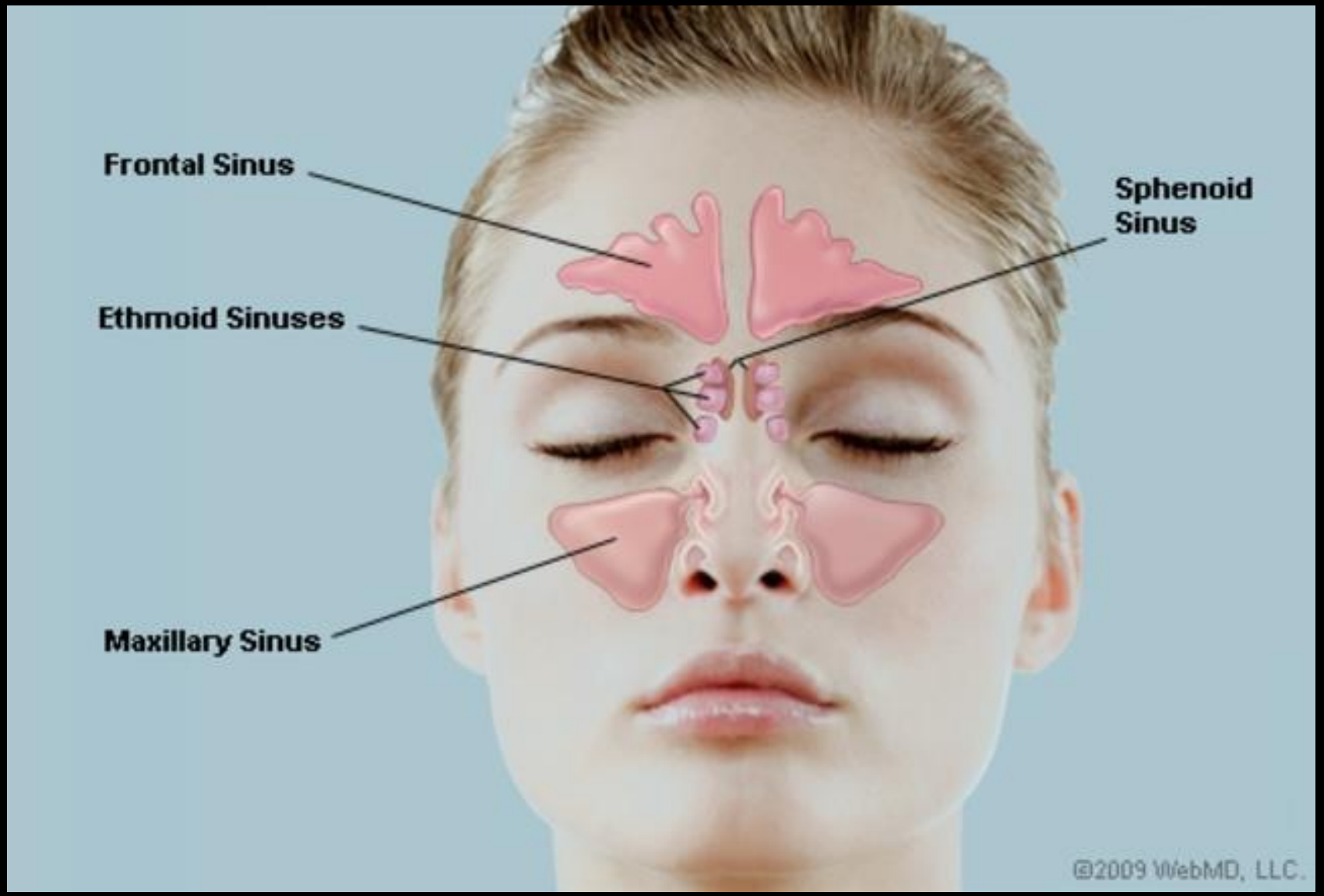
2-Size of electrode:

- -Uniform field → electrode is little larger than the part
- -Too small → superficial heat
- -Too big → heat loss
- -Unequal size → more heat concentrated under smaller electrode

FACTORS INFLUENCING THE FIELD:

- **3-Position of electrode relative to tissues**
- -Electrode parallel to the skin → Uniform field
- -Metal inside the tissue → concentration of field at metal as has least impedance and leads to burn
- -Air in cavities (uterus and sinuses) → field deviated to avoid it as air offers a high resistance, so the surface of cavity is only heated.

SINUS


Frontal Sinus
Sphenoid Sinus
Ethmoid Sinuses
Maxillary Sinus
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